

SUMEDH GAIKWAD

Backend Software Engineer — Full-Stack Capable, Learning AI/ML Integration
Pune, Maharashtra | +91 8208700967 | [Email](#) | [Portfolio](#) | [GitHub](#) | [LinkedIn](#) | [X](#)

SUMMARY

Backend-focused software engineer with working full-stack knowledge and hands-on experience building and deploying production backend and AI-driven systems. Published researcher in reinforcement learning applied to intelligent transportation systems. Completed a ~10-month full-stack development program and a software engineering internship with real industry exposure. Seeking backend / full-stack engineering roles to build systems that go beyond prototypes into real impact.

PROJECTS

Smart Traffic Management System — Final Year BE Project & SIH 2025 Hackathon Entry

Tech: Python, FastAPI, PyTorch, React, SUMO, TraCI, WebSocket

- Built a D-DQN reinforcement learning agent to dynamically optimize traffic signal phases in real time across two network topologies.
- Achieved 58.3% reduction in average waiting time and 43.8% speed improvement on a single intersection; 12.3% wait reduction on an urban arterial grid.
- Implemented dual parallel simulation for unbiased real-time RL vs. fixed-time comparison; integrated emergency vehicle detection with signal preemption.
- Deployed a live monitoring dashboard (FastAPI + React, WebSocket streaming); project selected for SIH 2025 and published as peer-reviewed research.

Atrio — Real-Time Collaborative Workspace

atrio-gamma.vercel.app

Tech: React, Node.js, Express, MongoDB, Socket.io

- Built a real-time collaboration platform for shared notes and task management with WebSocket-based live sync across users.
- Designed frictionless onboarding and deployed to real users during a beta phase.

Virtual2Reality — Real Estate Enquiry Website — Client Project

virtual2reality.in

Tech: Web Technologies, Lightweight Backend, Deployment

- Developed and deployed a production enquiry website end-to-end for a real estate firm in about a month, based on real client requirements.
- Built a lightweight backend capturing user enquiries into structured Excel records; delivered including deployment and client coordination.

PUBLICATION

Adaptive Urban Traffic Signal Optimization Using Reinforcement Learning: A D-DQN Approach

Prof. P.R. Patil, Sumedh Gaikwad, Nimish Darne, Yash Joshi | International Journal of Innovative Research in Technology (IJIRT), Vol. 12, Issue 11, pp. 14500–14513, April 2026, ISSN: 2349-6002

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, Java, C, C++, SQL

Frontend: React.js, HTML5, CSS3, Tailwind CSS, Responsive Web Design

Backend: Node.js, Express.js, FastAPI, RESTful APIs, WebSocket (Socket.io), Authentication (JWT), MVC Architecture

Databases & ORM: PostgreSQL, MongoDB, Prisma ORM

AI / Machine Learning: PyTorch, Reinforcement Learning, Machine Learning Fundamentals, Model Integration, Inference Pipelines

Tools & Platforms: Git, GitHub, Postman, Vercel, Render, VS Code, npm/pnpm, SUMO, TraCI

PROFESSIONAL EXPERIENCE

Software Engineering Intern — Actima Software, Pune

Jan 2025 – Aug 2025

- Developed and maintained frontend and backend features for multiple client projects, contributing to REST APIs, database integration, and production-ready web applications.
- Collaborated with clients to understand business requirements and translate feedback into functional software solutions across diverse service engagements.
- Worked closely with senior engineers in a lean development team, participating in Git-based workflows, code reviews, debugging, testing, and iterative software delivery.

EDUCATION

B.E. Computer Engineering — Honors: Artificial Intelligence & Machine Learning

Dec 2022 – June 2026

Savitribai Phule Pune University (SPPU) | CGPA: 8.33 (overall) | SGPA: 9.47 (latest semester)

CERTIFICATIONS & LEADERSHIP

Full Stack Web Development — upGrad

[View Certificate](#)

- Completed an intensive ~10-month program covering the MERN stack, Data Structures & Algorithms, and professional software development practices.

Smart India Hackathon 2025 — Team Lead, Team VEIG_1

- Led Team VEIG_1 through multiple screening rounds to college-level selection; built and presented the Smart Traffic Management System MVP.
- Project evolved into a final-year BE project and a peer-reviewed published paper — hackathon idea to publication.